Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION JUNE – 2025

Name of the Course: M. Tech

Semester: II

Name of the Paper: Modeling and Simulation

Paper Code: DEME 44

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the importance of mathematical modelling in dynamic system analysis.	CO1 & CO5
(b)	Derive the standard form of a second-order linear time-invariant (LTI) differential equation and discuss its significance.	
(c)	Explain the concept of damping ratio and its effect on system response? For a second-order system with $\zeta = 0.5$ and $\omega_n = 4$ rad/s, find the natural period and damped frequency?	
Q2	(20marks)	
(a)	Define and differentiate between complex numbers, complex variables, and complex functions with examples.	CO2
(b)	Explain the procedure of solving a linear time-invariant differential equation using Laplace transforms.	
(c)	Describe the use of Laplace transforms in modelling electrical and mechanical systems? Find the Laplace transform of $f(t) = te^{-2t}$	
Q3	(20marks)	
(a)	Define work, energy, and power in mechanical systems and discuss their relevance in system modelling?	CO2 & CO3
(b)	Discuss the use and importance of block diagrams in representing dynamic systems?	
(c)	What is the significance of the state-space model in control engineering? Convert $\ddot{x} + 3\dot{x} + 2x = u(t)$ into state-space form.	
Q4	(20marks)	
(a)	Explain the state-space representation of dynamic systems and its advantages over transfer function methods?	CO2 & CO3
(b)	Discuss the modelling of systems with and without input derivatives using the state-space approach?	
(c)	Explain the role of partial fraction expansion in control system analysis. Expand $\frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$ using partial fractions.	
Q5	(20marks)	
(a)	Explain how liquid level and pneumatic systems are mathematically modelled?	CO4
(b)	Discuss the process and necessity of linearizing a nonlinear dynamic system with an example?	

(c)	Explain the modelling of thermal systems and the significance of thermal resistance. Find the heat flow through a wall with thermal resistance of 5 K/W and temperature difference of 60°C?	